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DSAL Portfolio 01

Acknowledge, Describe, Evidence

Please append a completed copy of this form to your assessment submission.

Disclaimer:

The purpose of this form is to help you acknowledge and reference how you have used generative AI tools in your assessed work.

This form is ONLY for use with assessments for which written permission to use generative AI has been provided by the module convenor within the assessment instructions. If you are at all unsure what use is permitted for your module, please check with your module convenor.

The use of generative AI in the preparation of assessed work without written permission could be considered [academic misconduct](#).

Acknowledge:

I acknowledge the use of <insert AI system(s) and link> for the following purposes [tick as appropriate]:

- ☒ To generate materials for background research and self-study in the drafting of this assessment.
- ☒ To generate materials that were included within my final assessment in modified form.

Describe:

Provide a short summary of how you used generative AI in your assignment. You may wish to include the following information:

- The prompts you used
- The outputs you generated
- How you used/adapted/developed the outputs?

Add your short summary here:

I used generative AI (specifically GitHub Copilot and ChatGPT) as an assistive tool in this assignment, strictly following the guidelines provided in the assessment brief. My usage was limited to helping refine the Java code implementation after making an initial manual attempt, and to assist with structuring the explanation of concepts. I did not use AI to write any part of the written report, reflective narrative, or code comments.

Process Followed:

1. **Initial Attempt:** I first wrote a basic draft of the RPN calculator without AI assistance (see initial draft in Part B)
2. **AI Assistance:** Only after completing my initial attempt did I use AI to help improve the code structure and error handling
3. **Adaptation:** I adapted and modified all AI suggestions to fit the specific requirements of using the DSAL Stack ADT
4. **Documentation:** All AI interactions have been documented below

Evidence:

Please provide evidence of the outputs that you generated by copying and pasting below or by providing a screenshot. You should evidence:

- Generative AI system:
- Prompt:
- Output:

Provide your evidence here:

| **How I adapted it** | I took this structure but modified it to work with the specific DSAL Stack implementation from my module. I added the printStack() method for better trace visualization and enhanced the error messages to be more descriptive. |

Evidence 1: Trace Table Format

Generative AI system	ChatGPT 4
Prompt	"What's a good format for a trace table showing RPN expression evaluation step by step?"
Output	
	A trace table for RPN should show: Step number, Token being processed, Operation performed, Stack contents before, Stack contents after. Here's an example format:
	Step
	1
	2
	3
How I adapted it	I used this format but expanded it to include more detailed explanations in the Operation column and used consistent notation for stack representation (bottom → top).

Evidence 2: Error Handling Scenarios

Generative AI system	GitHub Copilot
Prompt	"What are the common error cases I should handle in an RPN calculator?"
Output	
	Common error cases to handle: 1. Division by zero 2. Insufficient operands for operator 3. Too many operands left after evaluation 4. Invalid tokens/characters 5. Empty expression 6. Stack underflow 7. Invalid operator usage
How I adapted it	I implemented specific error handling for each of these cases with descriptive error messages that help users understand what went wrong. I also added try-catch blocks specifically for the DSAL Stack operations which might throw exceptions.

Evidence 3: Testing Strategy

****Generative AI system****	ChatGPT 4
****Prompt****	"What test expressions should I use to thoroughly test an RPN calculator with + - * / ^ operations?"
****Output****	
	Recommended test expressions: 1. Simple: "5 3 +" (addition) 2. Multiple operations: "2 3 * 4 +" 3. Exponentiation: "2 3 ^" 4. Division: "10 3 /" (non-integer result) 5. Complex: "5 1 2 + 4 * + 3 -" 6. Error case: "5 0 /" (division by zero) 7. Error case: "5 +" (insufficient operands) 8. Nested: "3 5 2 ^ * 4 2 / +" 9. All operators: "4 2 / 3 5 * + 2 3 ^ -"
****How I adapted it****	I used these suggestions to create my test suite, adding a personalised expression (test 8) and ensuring all operators were tested. I also documented expected vs actual results in a table format.

Declaration

I confirm that:

- ☒ I made an initial attempt at the code without AI assistance
- ☒ AI was used only for assistive purposes as permitted in the assessment brief
- ☒ All AI outputs were adapted and modified to fit the specific requirements
- ☒ No AI was used to write any part of this written report or reflective narrative
- ☒ I have completed the University of Sussex AI Acknowledgement form

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****Date:** 20 Feb. 26

Declaration:

☒ I confirm that no content created by generative AI technologies has been presented as your own work.